

Sintra Inc. - Integrated Management System Blueprint

Comprehensive blueprint covering ISO 9001:2015, ISO 14001:2015, ISO 45001:2018, site safety, management protocols, and system integrator alliance strategies.

1. Integrated Management System (IMS) Framework

Sintra Inc. implements an Integrated Management System combining Quality (ISO 9001:2015), Environmental (ISO 14001:2015), and Occupational Health and Safety (ISO 45001:2018) management frameworks into a unified operation model tailored for automation and maintenance services.

Standard	Core Focus for Sintra Inc.	Key Output Records
ISO 9001:2015	Accuracy in PLC/SCADA programming, precision wiring, hardware verification, and zero-defect commissioning.	FAT/SAT records, Customer Satisfaction Surveys.
ISO 14001:2015	Safe collection, categorization, and ecological disposal of industrial electronic waste, old batteries, and wiring scrap.	E-Waste Log, Disposal Certificates.
ISO 45001:2018	Mitigation of live electrical shock hazards, lock-out tag-out (LOTO) protocols, and safe installation on cable trays.	Electrical Permits, Toolbox Talks Ledger.

2. Site Safety & Commissioning Protocols (Toolbox Talks)

Every operational morning, site managers must conduct a brief briefing (Toolbox Talk) with technical personnel before proceeding with execution.

Protocol A: Live Electrical Panel Safety & LOTO

- **Mandatory Check:** Always use calibrated Multimeters to check voltage absence prior to touching contacts.
- **Isolation:** Execute LOTO (Lock-Out Tag-Out) procedures on source circuit breakers; only the active technician holds the key.
- **PPE:** Class 00 / Class 0 insulated gloves and insulated safety shoes are strictly mandatory.

Protocol B: Safe Programming & On-Site Testing (PLC/SCADA)

- **Backup Verification:** Never push modified logic or code without saving a local time-stamped restore point.
- **Physical Clearance:** Before switching a PLC out of Stop/Safe mode to Run mode, ensure all mechanical areas are clear of personnel.
- **Dual-Loop Verification:** Signal loop checking must be synchronized strictly between the software desk and field technician via wireless communication.

3. Annual Internal Audit Schedule

To guarantee external audit success, Sintra Inc. executes a cross-departmental impartial audit scheme.

Department	Targeted Clauses (IMS)	Audit Focus Points
Engineering & Design	ISO 9001: Clause 8.3	Software revision control, backup logs, design inputs validation.
Site Operations	ISO 45001 / ISO 14001	Execution of LOTO logs, hazardous e-waste containment, PPE adherence.
Procurement / Stores	ISO 9001: Clause 8.4	Vendor evaluations, calibration validity of testing equipment.

4. Management Review Agenda

Annual executive meeting to verify the structural health and compliance of Sintra Inc.'s operational framework.

- **Input Parameters:** Previous audit non-conformities, external project feedback, resource availability, and process key performance indicators (KPIs).
- **Decision Outputs:** Resource allocations, optimization modifications to infrastructure, and targeted training pipelines for engineering teams.

5. Global Vendor Alliance Strategy (Siemens / Schneider Electric)

The strategic path to achieve official System Integrator status and accelerate commercial acquisition velocity:

1. **Talent Pipeline Certification:** Enroll minimum 2 engineers in official certification pipelines (Siemens TIA Portal ST-PRO or Schneider EcoStruxure Expert).
2. **Track Record Compilation:** Document 3 successful automation case studies showcasing proper wiring, programming standards, and clean electrical panel assembly.
3. **Compliance Sync:** Leverage this ISO-based documentation framework to satisfy the rigorous business management requirements demanded during vendor audits.

Disclaimer: This system documentation blueprint is compiled for corporate structural development purposes based on ISO guidance and standard engineering practices. All field testing and execution must strictly respect active country laws and engineering safety declarations.